

Expanded Research Plan: Spectral Diagnostic Trajectories (SDT)

Executive Summary

This document outlines the next evolutionary phase of the Geodesic Diagnostic Trajectories (GDT) framework. While GDT successfully models clinical intent as a continuous trajectory on the Riemannian manifold of Symmetric Positive Definite (SPD) matrices $\mathcal{S}_d^+$, its reliance on sliding-window covariance bounds its ability to natively ingest highly unstructured, continuous, and multi-modal physiological streams.

The **Spectral Diagnostic Trajectories (SDT)** framework transitions the state space from a Riemannian manifold to a dynamic multi-modal hypergraph. It replaces the geodesic shift ratio κ_t with spectral gap analysis (Fiedler value λ_2) for pivot detection, and replaces JEPA-driven latent prefetching with spectrally-biased random walks over the global EHR graph.

Phase 1: Hypergraph Construction & Multimodal State Space

Objective: Encode the physician's cognitive path and the patient's physical state into a unified, continuous hypergraph topology, building upon the differential geometry established in GDT.

- **Node Initialization and Embedding Formulation:**
 - The raw query or clinical action initiated by a clinician at turn t is denoted as q_t .
 - **Textual & Semantic Nodes:** Ingest clinical notes, query logs, and discrete EHR events. These are processed into a dense embedding $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ using models like ClinicalBERT. To maintain computational efficiency, the original 768-dimensional embeddings are projected down to a 128-dimensional subspace using a learned linear layer, preserving the semantic geometry established in GDT.
 - **Physiological Nodes:** Expand the state space to ingest continuous data (e.g., MIMIC-IV PPG-ECG waveforms, HiRID continuous vitals). These nodes represent high-frequency physiological states encoded via 1D-CNN or state-space models (e.g., Mamba).
- **Covariance Elicitation and Baseline State Tracking:**
 - As a foundational bridge from GDT, the local semantic intent at turn t is initially represented by a symmetric positive definite (SPD) matrix computed over a

sliding window:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^{\top} + \lambda I_d$$

where k is the sliding window size, μ_t represents the local mean of the embeddings, and λI_d strictly guarantees positive definiteness. 0 is a small regularization scalar) strictly guarantees numerical stability and positive definiteness." type="suggestion">

- The state matrix S_t resides in the Riemannian manifold of SPD matrices $\mathcal{S}_{d^+}$, equipped with the affine-invariant metric tensor:
 $g_S(u, v) = \text{Tr}(S^{-1} u S^{-1} v)$
- The evolving diagnostic uncertainty is quantified through the differential entropy of the Gaussian density:
 $H(S_t) = \frac{1}{2} \log \det(2\pi e S_t)$

- **Evolution to Hyperedge Formation:**

- While GDT relies entirely on the covariance matrix S_t , SDT transitions this into a dynamic active subgraph $G_t = (V_t, E_t)$ to accommodate the multimodal nodes.
- Unlike GDT's pairwise covariance, hyperedges can connect multiple nodes simultaneously.
- Edges are weighted based on a combination of the clinician's temporal clickstream sequences (sequential proximity) and cosine similarity in the latent embedding space.
- The instantaneous diagnostic hypothesis is thus continuously tracked as this evolving active subgraph G_t , expanding upon the foundational Riemannian state tracking.

Phase 2: Pivot Detection & The Interference Gate (GDT to SDT Evolution)

Objective: Mathematically detect abrupt changes in diagnostic intent to trigger the interference gate, transitioning from GDT's differential geometry to SDT's spectral graph theory.

The Legacy GDT Geodesic Trigger

In the foundational GDT model, sudden cognitive pivots (e.g., a clinician shifting focus from respiratory distress to acute kidney injury) were detected by measuring the acceleration of the state matrix S_t across the Riemannian manifold.

- **Geodesic Distance:** The distance between consecutive states is measured as:
 $d_g(S_{t-1}, S_t) = \sqrt{\log(S_{t-1}^{-1/2} S_t S_{t-1}^{-1/2})}$
- **The Shift Ratio (κ_t):** To differentiate between normal evidence accumulation and an abrupt pivot, GDT calculates the ratio of the current step to the historical moving

average:

$$\kappa_t = \frac{d_g(S_{t-1}, S_t)}{\frac{1}{k} \sum_{i=1}^k d_g(S_{t-i-1}, S_{t-i})}$$

- **Interference Gate Activation:** If $\kappa_t > \kappa_0$ (where κ_0 is a tunable threshold), the system identifies a "contextual breach." The interference gate triggers, freezing the stale context to prevent it from polluting the new diagnostic trajectory.

The SDT Spectral Trigger Evolution

While the geodesic shift ratio κ_t is highly effective for discrete semantic embeddings, it becomes computationally unstable when applied to continuous, high-frequency physiological nodes. SDT resolves this by mapping the pivot detection to the eigenvalues of the evolving hypergraph G_t .

- **Laplacian Monitoring:** At each time step t , compute the normalized graph Laplacian matrix for the active subgraph G_t :
$$L_t = I - D_t^{-1/2} A_t D_t^{-1/2}$$
where A_t is the adjacency matrix and D_t is the degree matrix.
- **Fiedler Value Tracking (λ_2):** Monitor the spectrum of L_t , specifically the second smallest eigenvalue, known as the Fiedler value $\lambda_2(L_t)$.
 - **Diagnostic Cohesion:** λ_2 strictly quantifies the algebraic connectivity of the diagnostic hypothesis. Steady growth in λ_2 represents the clinician gathering corroborating evidence (nodes are densely connected, uncertainty drops).
- **Spectral Interference Gating:** A sudden, precipitous drop in λ_2 mathematically indicates that the active subgraph is fracturing into distinct clusters—representing a clinician maintaining stale nodes (the old hypothesis) while simultaneously querying entirely new nodes (the new hypothesis).
 - **New Trigger:** The interference gate now triggers when the spectral gap expands rapidly:
$$\Delta \lambda_2 > \tau_{\text{pivot}}$$
 - This functional replacement of $\kappa_t > \kappa_0$ allows the system to detect cognitive pivots across multi-modal data structures in near real-time, purely through graph topology.

Phase 3: Graph-Signal Context Attenuation

Objective: Bound context contamination and prevent stale hypotheses from polluting downstream trajectory simulation by executing a mathematically rigorous Fiedler cut.

- **Spectral Filtering:**
 - Upon trigger activation ($\Delta \lambda_2 > \tau_{\text{pivot}}$), the system applies a high-pass graph filter to the node signals. Let x represent the signal of contextual relevance across the nodes. The filtered signal is $\tilde{x} = H(L_t)x$,

where the filter H actively suppresses low-frequency structural components (which heavily correlate with the pre-pivot stale context).

- **Subgraph Decoupling (The Fiedler Cut):**
 - When the spectral gap triggers, the system isolates the structural fracture using the Fiedler vector v_2 (the eigenvector associated with the Fiedler value λ_2).
 - The entries of v_2 assign a continuous value to each node in the active subgraph G_t .
 - **Partitioning:** The graph is mathematically partitioned into two disjoint sets based on the algebraic sign of the Fiedler vector components:

$$V_{\text{stale}} = \{i \in V_t \mid v_2(i) < 0\}$$

$$V_{\text{active}} = \{i \in V_t \mid v_2(i) \geq 0\}$$
 - **Severing Edges:** All hyperedges connecting V_{stale} (the old diagnostic trajectory) and V_{active} (the newly forming hypothesis) are systematically severed. This enforces a strict, instantaneous bound on information interference, directly advancing GDT's Theorem 3.2 (Curvature-Gated Contamination Decay) from Riemannian geometry into discrete graph space.
- **Embedding Formulation:**
 - Once the stale subgraph is decoupled, the system aggregates only the nodes in V_{active} to generate a cleanly filtered, highly focused retrieval vector.
 - This isolated vector serves as the seed for generating the next immediate query state, completely free of latent-context pollution from the prior diagnostic path.

Phase 4: Anticipatory Prefetch Engine & Simulation Interface

Objective: Evolve the retrieval engine into a predictive world model driven by optimal control constraints.

- **Optimal Control Formulation:**
 - Similar to GDT, prefetching is framed mathematically to minimize expected clinical decision delay, operating natively on graph topologies. Let u_t denote the control action (the subset of nodes selected for prefetching from the global graph).
 - The system optimizes a continuous loss function $J(u_t)$ that balances temporal retrieval delay, structural relevance, and cognitive load bounds:

$$J(u_t) = \mathbb{E}[\alpha \mathcal{L}_{\text{delay}}(G_{t+1}, u_t) - \beta \mathcal{R}_{\text{struct}}(u_t) + \gamma \Omega(u_t)]$$

Where $\mathcal{L}_{\text{delay}}$ calculates the time saved by having nodes readily available, $\mathcal{R}_{\text{struct}}$ rewards structural similarity to the active hypothesis, and $\Omega(u_t)$ penalizes overwhelming the clinician's cognitive bandwidth.
- **Biased Traversal (Spectrally-Biased Random Walk):**

- Replace the JEPA-based target network with a parameterized random walk on the global EHR graph G_{global} .
- Let standard transition probabilities be $P_0(v_j \rightarrow v_i) = \frac{A_{ij}}{D_{ii}}$. The system instead computes a *spectrally-biased* transition probability, mapping nodes into a spectral embedding space $\phi(v)$ to heavily weight trajectories structurally similar to the active subgraph:

$$P(v_j \rightarrow v_i) = \frac{P_0(v_j \rightarrow v_i) \exp(-\tau \|\phi(v_i) - \phi(v_j)\|_2^2)}{\sum_{k \in \mathcal{N}(i)} P_0(v_k \rightarrow v_i) \exp(-\tau \|\phi(v_i) - \phi(v_k)\|_2^2)}$$
- This spectral biasing mechanism allows the system to simulate branching physiological futures and natively forecast the next clinical intent across multi-modal nodes.
- **Cognitive Gating:**
 - To adopt the cognitive safety principles of GDT (mitigating premature closure and automation bias), the prefetch volume is mathematically restricted.
 - Let $H(v \rightarrow G_t)$ represent the cognitive load increment of presenting node v relative to the current active context. The optimal control action u_t is strictly subjected to the human cognitive-memory constraint (informed by Sweller's cognitive-load theory):

$$\sum_{v \in u_t} H(v \rightarrow G_t) \leq C_{\text{max}}$$
 - This directly prevents UI overwhelming and ensures the human-in-the-loop retains full agency over the diagnostic trajectory.

Phase 5: Evaluation & Operational Analytics (Next Steps)

Objective: Validate SDT using the established benchmark methodology from the GDT study, and quantify the downstream operational impact on hospital throughput.

- **Data Sources:** Re-use the multi-domain MIMIC-IV, MIMIC-CXR, and HiRID datasets, but actively integrate the continuous MIMIC-IV PPG-ECG waveform metadata for the multimodal node initialization (Phase 1).
- **Crossover Benchmark Metrics:** Conduct a randomized crossover evaluation comparing SDT against the original GDT framework, CMA, and BM25. Primary metrics include:
 - **Time-to-Correct-Information (TTCI):** The cumulative temporal delay before the target diagnostic node is natively fetched and displayed.
 - **Mean Reciprocal Rank (MRR):** Evaluates the rank of the first relevant node fetched immediately post-pivot.
 - **Contextual Precision:** The ratio of active nodes (V_{active}) correctly retained versus stale nodes (V_{stale}) inappropriately bleeding into the new trajectory post-Fiedler cut.
- **Queueing Model Translation (\$M/M/c\$):** To measure system-wide impact, the clinical time saved by SDT is mapped into an Erlang C queueing model representing Emergency Department (ED) operations.

- Let λ be the patient arrival rate and c be the number of active clinicians. Let the baseline average clinical diagnostic service time be $\frac{1}{\mu}$.
- SDT accelerates information retrieval, saving an average duration of Δt per patient. The new accelerated service rate becomes:

$$\mu' = \frac{1}{\frac{1}{\mu} - \Delta t}$$
- The ED traffic intensity updates to $\rho' = \frac{\lambda}{c\mu'}$.
- The probability of a patient waiting for a bed (using the Erlang C formula, P_W) is calculated as:

$$P_W = \frac{\frac{(c\rho')^c}{c!} \frac{1}{1-\rho'}}{\sum_{k=0}^{c-1} \frac{(c\rho')^k}{k!} + \frac{(c\rho')^c}{c!} \frac{1}{1-\rho'}}$$
- **Impact on Wait Time (W_q):** The projected reduction in patient queueing wait time is evaluated as:

$$W_{q'} = \frac{P_W \cdot (\mu')^{-1}}{c(1 - \rho')}$$
- This mathematical translation proves that even marginal reductions in cognitive retrieval latency (Δt) driven by the anticipatory prefetch engine yield non-linear, exponential reductions in ED crowding and patient length-of-stay (LOS).